SMART EXPENSE TRACKER

A MINI C++ PROJECT

TEAM MEMBERS

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ITRODUCTION

- Expense management is important in daily life
- Manual tracking is difficult and error-prone
- This project provides a simple digital solution
- Built using C++ and basic programming concepts

OBJECTIVES

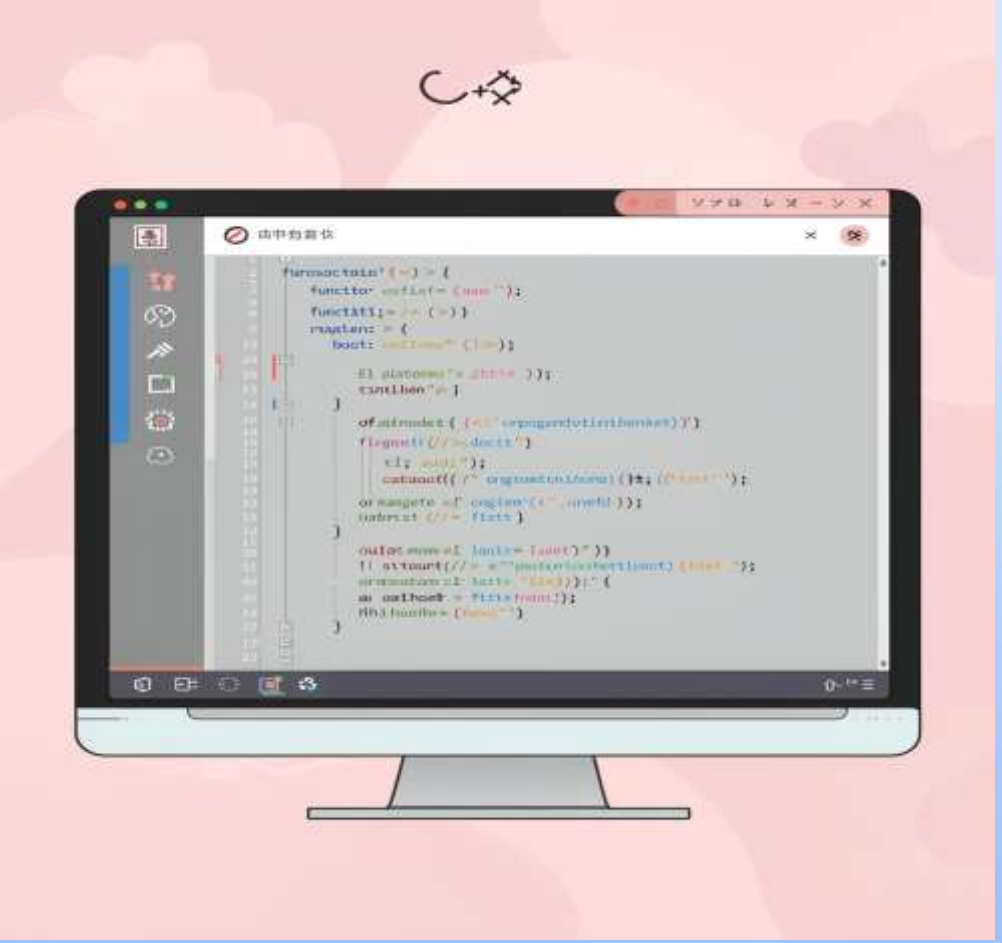
- ❖ To develop a system to manage expenses
- ❖ To allow users to:
  - ✓ Add expenses
  - ✓ View records
  - ✓ Search data
  - ✓ Delete entries
- ❖ To implement using C++ and data structures

FEATURES

- ✓ 🔐 Login system
- ✓ ➕ Add expense
- ✓ 📄 View expenses
- ✓ 🔍 Search by date
- ✓ 📅 Monthly report
- ✓ 💰 Total expense
- ✓ ❓ Delete expense

TECH NOLOGIES USED

- ❖ Programming Language: C++
- ❖ Concepts Used:
  - ➢ Structures
  - ➢ Vectors
  - ➢ Loops
  - ➢ Functions
  - ➢ Exception Handling



METHODOLOGY

- Menu-driven program
- Data stored using vector (dynamic array)
- User input handled using
- getline()
- Operations performed:
  - Insert
  - Delete
  - Search
  - Display

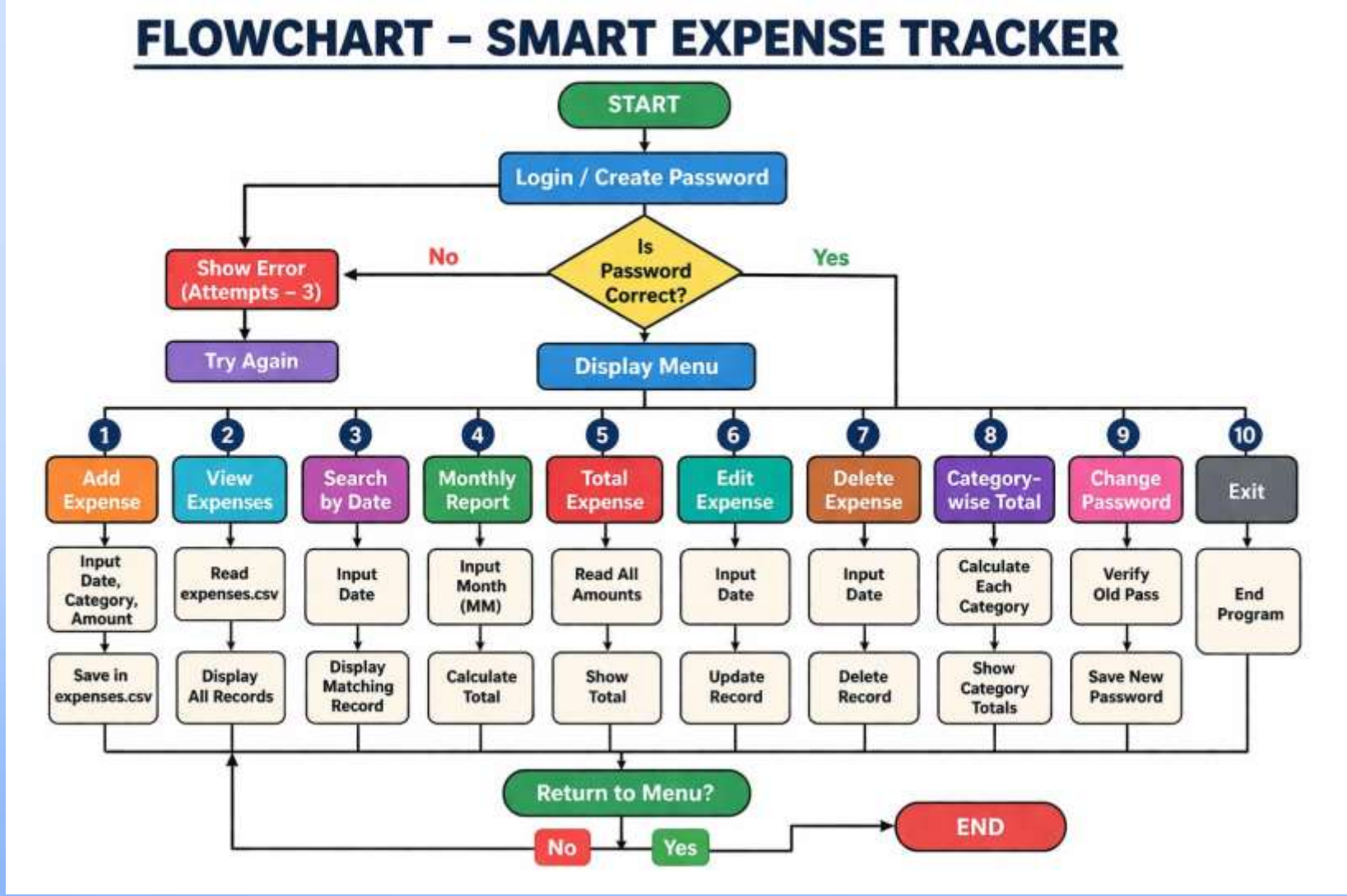
System Workflow

- Start program
- Create & verify password
- Display menu
- User selects operation
- Perform action
- Repeat until exit

ALGORITHM

- Start
- Login authentication
- Display menu
- Take user input
- Perform selected operation
- Repeat until exit
- End

FLOWCHART


FLOWCHART – SMART EXPENSE TRACKER
START
Login / Create Password
Is Password Correct?
No
Yes
Show Error (Attempts – 3)
Try Again
Display Menu
1
2
3
4
5
6
7
8
9
10
Add Expense
View Expenses
Search by Date
Monthly Report
Total Expense
Edit Expense
Delete Expense
Category-wise Total
Change Password
Exit
Input Date, Category, Amount
Read expenses.csv
Input Date
Input Month (MM)
Read All Amounts
Input Date
Input Date
Calculate Each Category
Verify Old Pass
End Program
Save in expenses.csv
Display All Records
Display Matching Record
Calculate Total
Show Total
Update Record
Delete Record
Show Category Totals
Save New Password
Return to Menu?
No
Yes
END

ADVANTAGES

- ✓ Simple and easy to use
- ✓ Fast execution
- ✓ No external dependencies
- ✓ Error handling included
- ✓ Efficient data management

FUTURE ENHANCEMENTS

- File storage (save data permanently)
- GUI interface
- Edit expense feature
- Charts and analytics
- Mobile/web version

LIMITATIONS

- Data is not stored permanently
- No graphical interface
- Works only in console

CONCLUSION

- Successfully implemented expense tracker
- Demonstrated use of C++ concepts
- Efficient for small-scale usage Can be expanded into real-world application

Thank You